Public Perception, Health Impact, and Behavioral Responses to Air Pollution

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***Abstract*—Air pollution has become a pressing global issue, significantly affecting public health and quality of life, particularly in rapidly urbanizing and industrializing regions. This study explores public perceptions of air quality and its associated health risks, with a specific focus on highly polluted cities like Lahore and Delhi. The research identifies critical factors influencing these perceptions, including age, respiratory health history, education level, and proximity to polluted areas. Using a range of statistical methods such as chi-square, Kruskal-Wallis, ANOVA, and Wilcoxon Rank-Sum tests, the study analyzes the relationships between the Air Quality Index (AQI), health symptoms, and demographic variables. Results indicate a strong correlation between AQI levels and their impacts on daily activities, such as work and study, and between precautions taken and health outcomes. However, variations in air quality perceptions across genders and professions were found to be minimal.**

The study also highlights a significant gap in public understanding of AQI values and their implications, which often leads to inadequate protective measures. Despite growing awareness of air pollution's health effects, many individuals remain unaware of the specific risks posed by moderate AQI levels or fail to adopt effective preventive behaviors. Behavioral adaptations, such as avoiding outdoor activities during high-pollution periods and using air purifiers, reflect an increasing awareness of pollution risks, yet underscore the urgent need for improved public education and communication strategies.

Additionally, public opinions on government efforts to address air pollution showed consistency across genders, suggesting that environmental campaigns and policies could be equally impactful for diverse demographic groups. The study emphasizes the importance of combining stricter environmental regulations with targeted public awareness campaigns to promote sustainable behaviors and improve air quality. These findings provide actionable insights for policymakers and highlight the need for collaborative regional and global efforts to tackle air pollution and its health consequences effectively.

I. Introduction

Air quality plays a crucial role in moulding public health and overall lifestyle. With increase in urbanization and industriallization air pollution has become a global concern. People's awareness and perceptions of air pollution and its effects on health. The research emphasizes the need for standardized and globally reliable measures [4]. This paper aims to point out the increasing level of smog in Lahore and Delhi and its causes, severe air pollution was observed in Lahore and Delhi during the fifth season due to crop residue burning and meteorological conditions, which negatively affects health and the economy and addresses air pollution issues with regional cooperation and ground monitoring [9]. It examines the health effects of winter fog on different occupational groups in Lahore and surrounding areas. This study highlights key risk factors such as age, history of respiratory disease, education, and living in polluted areas, and highlights the socio-economic impact on respiratory diseases through survey and logistic regression [8]. It gives insights to international level by exploring the air quality and pollution perception of the residents living in China.The results show that both pollution perceptions and actual pollution have a negative impact on mental health, highlighting the need to reduce environmental health risks [13]. This study maps the gap between actual level of air pollution and public perception by taking into count people's daily movement. The results show that people with lower incomes are more exposed to pollution, while students and older people underestimate air pollution,

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which provides important guidance for policymakers [11]. It examines perceptions of air quality and health concerns during long-term air pollution, wildfires, and COVID-19 lockdowns. The results show that these factors have a significant impact on public health and awareness of pollution [6].This study highlights the worsening smog situation in three cities of Pakistan. It also highlights the increasing environmental and health impacts of smog [2].

II. Previous Work

There are few studies that have also put efforts to wrap up information on this topic. Some papers and their conclusion are stated below: Despite growing awareness of air pollution as a global health concern, limited research has been conducted on methods for evaluating how individuals perceive these risks. This scoping review sought to examine existing studies that assess people's perceptions of air pollution and its impact on health, while also exploring approaches used to measure these perceptions as an essential aspect of health-related behavior. The review followed the methodological framework proposed by Arksey and O'Malley. PubMed and Web of Science were searched [4]. Social, geographical, and contextual factors heavily influence perceptions of indoor air pollution and should inform public health communication strategies. Our findings highlight that common practices to improve air quality can pose risks, emphasizing the need for tailored, context-aware risk communication. This study adopted a mixed-methods approach which used a guided questionnaire designed to elicit both quantitative and qualitative data, collected as digital voice notes [12]. We have also found some research work related to pollution in South Asia, this study explorestrends in air quality, pollution, and sustainability in South Asia from a population-based perspective. It highlights environmental challenges in the region and their impacts on public health and sustainability [1]. Another study examined the perceptions of air pollution among adults commuting to work in Queensland, Australia. The study collected information about travel patterns and risks through a questionnaire [3]. We have also found some working on the environmental and health impacts of air pollution.This paper examines the environmental and health impacts of air pollution. It sheds light on the impact of air pollution on public health and its various environmental aspects [10]. There is also a research that examines measures and methods for assessing public perception of air pollution and its health impacts. It covers different ways of understanding the risks of air pollution [4]. Another study explores the impact of air quality and pollution perception on health in China. Perceived pollution harms self-rated and mental health, while objective pollution mainly affects mental health, highlighting the need for targeted environmental policies [13]. This study highlights mismatches between perceived and actual air pollution, influenced by factors like income, age, mental health, and environment. Vulnerable groups are often more accurate, guiding policymakers to enhance awareness and health-conscious planning [11]. This study examines the link between perceived and objective air pollution and respiratory health in New York City, finding that perceived air quality is more strongly associated with asthma than actual pollution levels. Mental health factors were adjusted for, but no significant association was found with general health. These results suggest perceived air quality plays a significant role in asthma development [5].

III. Methodology

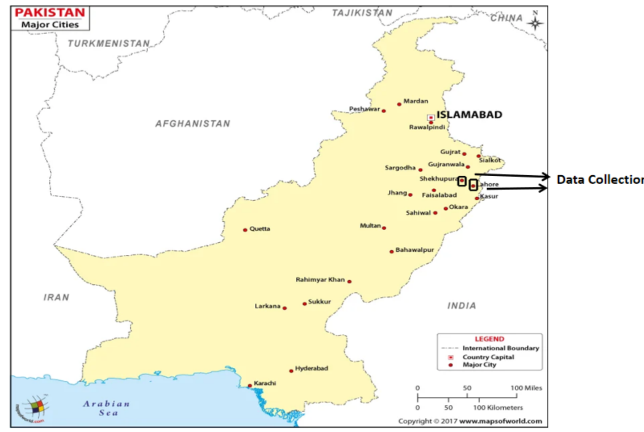

Fig. 1: Collection of Data

The methodology for data collection in this study combined both physical and digital approaches to ensure a broad and diverse representation of participants. Physical surveys were conducted in Lahore and Sheikhupura, complemented by online surveys to capture a wider demographic. The sample population consisted of approximately 60.9 % males and 30.1% females, in fig 2.

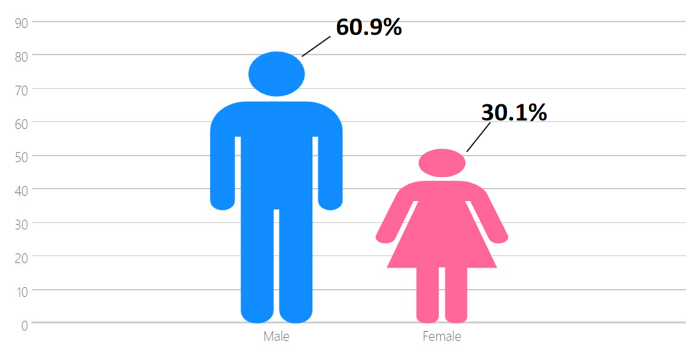

Fig. 2: Distribution of Data

In our research paper, we use Chi-square test to examine the association between gender and people's opinions, the relationship between AQI and its impact on study and work, the AQI rating between people as shown in figure 3 and whether precautions are associated with symptoms as shown in figure 4 and 5. We have also applied another test named Kruskal-Wallis and we use this test to calculate the median between people

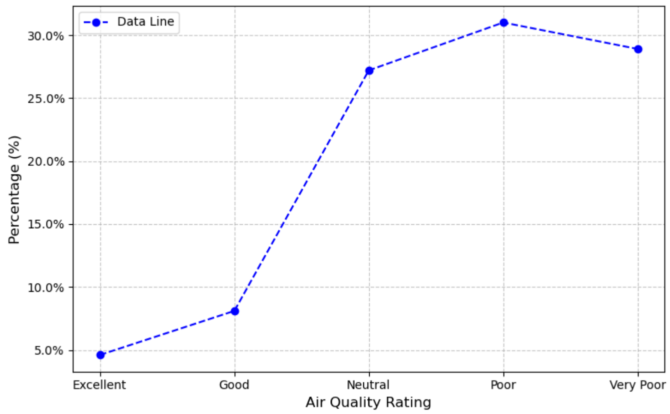

Fig. 3: AQI Rating

of different professions and the status of air quality in their respective area over past five years. We have also managed to apply Annova which we used to calculate the mean between two or more groups. Here we have applied it to calculate the mean between air quality ratings by people with different professions. Moreover, we have applied Wilcoxon Rank-Sum stats to calculate the difference between gender opinion about government performance for reducing air pollution and the air quality ratings and how do it connects with the gender opinion. Lastly, we have applied the Runs test for randomness on the feature whether people know the term AQI or not to verify that whether our dataset is random or not.

IV. Experiment and Results

First, we apply Chi-Sqaure

$$\chi^2_{\text{cal}} = \sum \frac{(O_f - E_f)^2}{E_f}$$

Where O_f is the observed frequency,
and E_f is the expected frequency.

For the chi-square experiment we developed our null hypothesis as no association between gender and people opinion and alternative hypothesis as there is association between gender and people opinion. From these hypothesis this study aims to calculate the association between gender and people opinion and we came to know that there is no association between them. Another null hypothesis we formed was there is no association between AQI and its impact on study and work and alternate hypothesis as there is association between AQI and its impact on work and the result came out to be that there is significance association between them. Another null hypothesis to be tested was there is no association between precautions and symptoms and the alternate hypothesis that we formed was there is association between precautions and symptoms. The aim to form these hypothesis was to calculate association between precautions and symptoms and to find the relationship between them. And we find the result to be in the favor of highly association between the two as shown in figure 3 and 4.

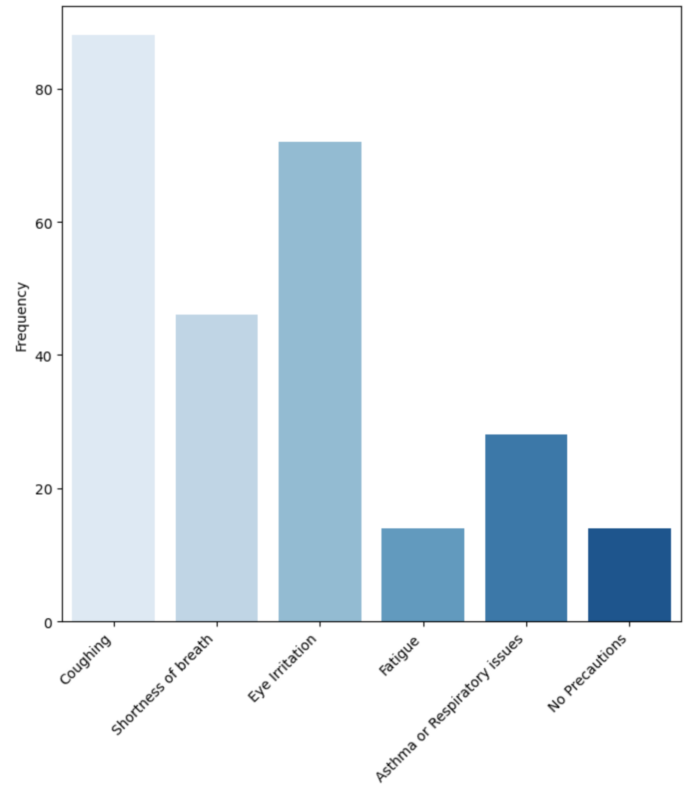

Fig. 4: Distribution of Symptoms

We apply also Kruskal-Wallis test

$$H = \frac{12}{N(N+1)} \left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \cdots + \frac{R_k^2}{n_k} \right) - 3(N+1)$$

where

R_1 = sum of ranks for Sample 1
n_1 = number of observations in Sample 1
R_2 = sum of ranks for Sample 2
n_2 = number of observations in Sample 2
R_3 = sum of ranks for Sample 3
n_3 = number of observations in Sample 3
N = total number of observations in all samples combined
$N = n_1 + n_2 + n_3$

which we developed our null hypothesis as there is no difference in opinion and people of different profession on air quality in their respective areas over the past five years and the alternate hypothesis goes with the difference in opinion and people of different professions on air quality in their respective areas over the past five years. And we have concluded that there is no difference in opinion and people of different

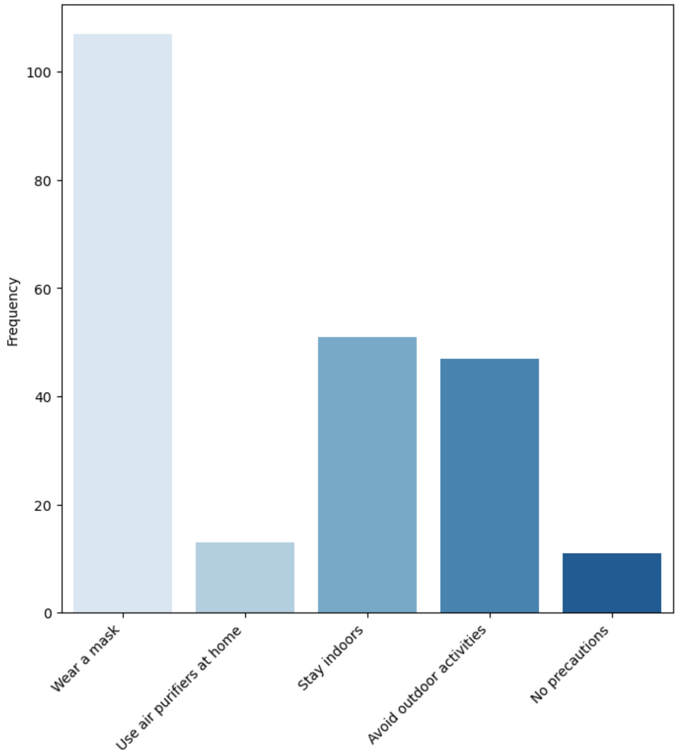

Fig. 5: Distribution of Precautions

profession on air quality in their respective areas over the past five years as shown in fig 6. Another test that we applied is

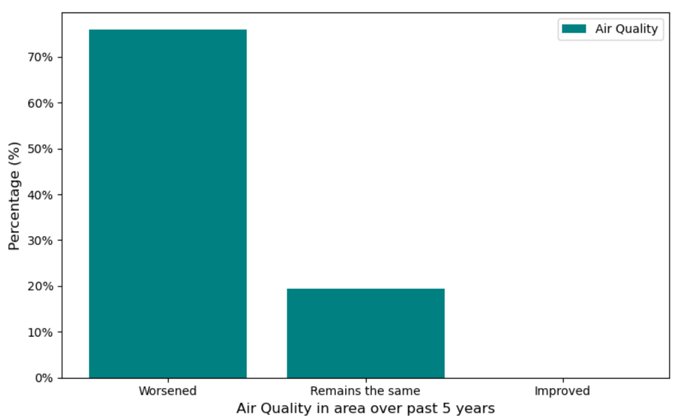

Fig. 6: Air Pollution in area over past 5 years

Annova

$$SSA = n\sum_{i=1}^{k}(\bar{y}_{i.} - \bar{y}_{..})^2$$

$$SSE = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})^2$$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})^2$$

$$s_1^2 = \frac{SSA}{k-1}, \quad s^2 = \frac{SSE}{k(n-1)}$$

$$f = \frac{s_1^2}{s_2^2}$$

for which we have calculated our null hypothesis that the mean air quality are same across different professions and the alternate hypothesis as atleast one profession has different mean.And we have came to know that mean of air quality across different profession is same as shown in fig 7.

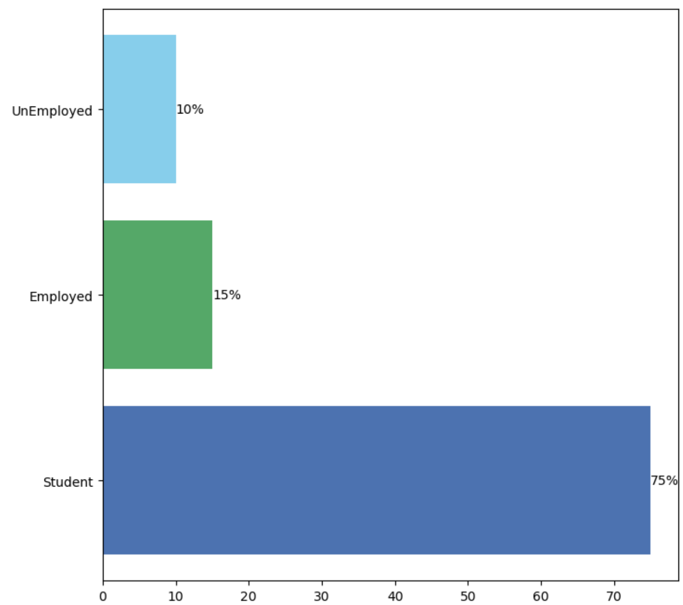

Fig. 7: People Opinion about Government Measures

We have also applied Wilcoxon-Rank Sum statistics

$$z = \frac{R - \mu_R}{\sigma_R}$$

Where

$$\mu_R = \frac{n_1(n_1 + n_2 + 1)}{2}$$

$$\sigma_R = \sqrt{\frac{n_1 n_2(n_1 + n_2 + 1)}{12}}$$

$$n_1 = \text{size of the sample from which the rank sum } R \text{ is found}$$

$$n_2 = \text{size of the other sample}$$

$$R = \text{sum of ranks for the sample with size } n_1$$

and stated the null hypothesis as no difference between gender opinion about government performance in reducing air pollution and alternate hypothesis as there is significance difference between gender opinion about government performance in reducing air pollution. And the result has came to known in the favor of no difference between gender opinion about government performance in reducing air pollution as shown in fig 8. And lastly we applied Random Runs Test

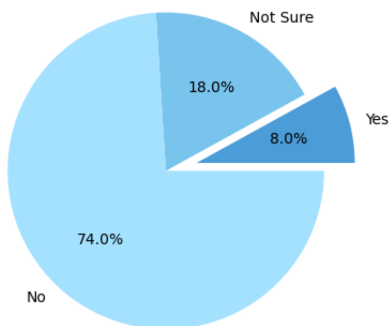

Fig. 8: People Opinion about Government Measures

$$z = \frac{G - \mu_G}{\sigma_G}$$

Where

$$\mu_G = \frac{2n_1n_2}{n_1 + n_2} + 1$$

and

$$\sigma_G = \sqrt{\frac{(2n_1n_2)(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}}$$

to verify that our dataset is random and we came to know that it is actually random.

As we see from the fig 10, According to the people the most dangerous cause are industrial pollution and vehicle emissions as it was also observed during Covid-19 that the pollution level was considerably decreased due to fewer industrial activities [7].

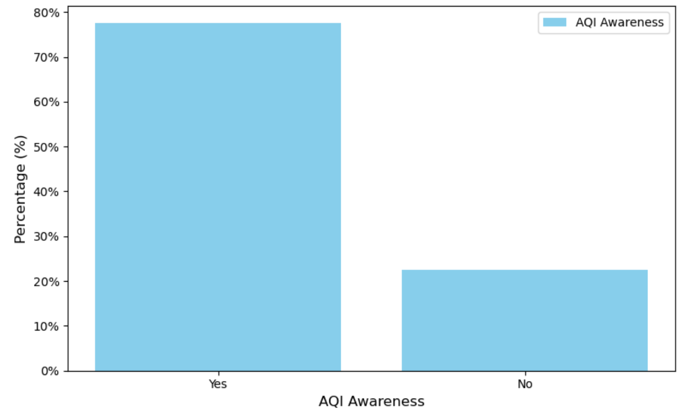

Fig. 9: AQI Awareness

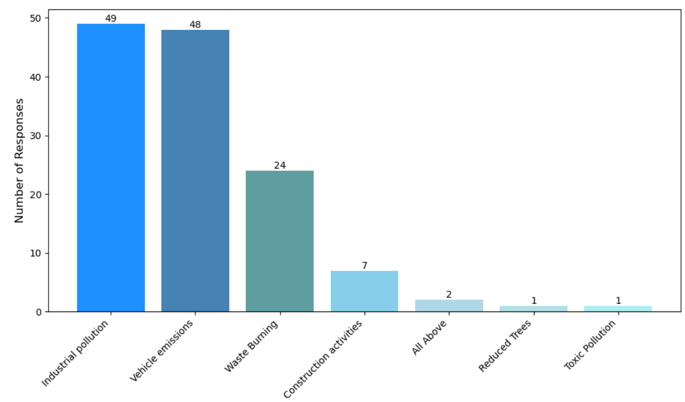

Fig. 10: Main cause of air pollution

V. Conclusion

This study examined various dimensions of air quality perception and its health impacts, focusing on the relationships between gender, the Air Quality Index (AQI), precautions, and symptoms. The chi-square test revealed that there is no significant link between gender and public opinions, but a noteworthy association was found between AQI and their effects on work and study. A strong correlation was also identified between precautions taken and symptoms experienced.

Further statistical analysis, including Kruskal-Wallis and ANOVA tests, indicated no substantial differences in air quality perceptions across different professions, and the mean air quality ratings were consistent across professional groups. The Wilcoxon Rank-Sum test revealed that there is no significant gender-based differences in opinions regarding government actions to tackle air pollution.

The Runs test for randomness confirmed that the dataset was indeed random. These findings suggest that while perceptions of air quality and individual precautions significantly influence the understanding of pollution's health effects, the role of gender and profession appears minimal. The study underscores the importance of tailored public awareness campaigns and effective policies to mitigate the health impacts of air pollution, particularly in regions facing significant environmental challenges.

Air pollution poses serious health risks, particularly for individuals who are less cautious or unaware of the potential dangers of poor air quality. Prolonged exposure to pollution is associated with respiratory, cardiovascular, and mental health issues. People living near high-pollution areas, such as industrial zones or highways, are especially vulnerable. Those who fail to take protective measures, such as wearing masks or staying indoors when pollution level is high, are on high risk of experiencing the negative health effects of air pollution. Thus, raising awareness and encouraging protective behaviors are critical to minimizing the health risks associated with air pollution.

Although many are aware of the dangers of air pollution, many still lack a complete understanding of the AQI and its health implications. The AQI is a vital tool for monitoring air quality, measuring pollutants like particulate matter (PM2.5), nitrogen dioxide, sulfur dioxide, carbon monoxide, and ozone. However, many people don't fully understand the AQI values or the health risks they represent, which can lead to misunderstandings about when precautions should be taken. For example, people may underestimate the health hazards of moderate AQI levels or may not know the actions needed to protect themselves during periods of high pollution. Educating the public on how to interpret AQI values and their health implications is essential for empowering individuals to make informed decisions and safeguard their health.

Public opinions on government efforts to reduce pollution seem consistent across genders, with both men and women showing similar views on the effectiveness of policies aimed at addressing air pollution. This indicates that gender does not significantly influence opinions on government actions concerning environmental issues. Despite individual differences on how people prioritize environmental concerns, there appears to be a shared concern for improving air quality and reducing pollution. This finding is valuable for policymakers, as it suggests that environmental campaigns and initiatives may be equally effective across gender groups, allowing for more inclusive strategies.

When faced with poor air quality, many people adjust their behaviors to minimize exposure, such as avoiding outdoor activities during high pollution periods. This is a common strategy to protect oneself from the harmful effects of pollutants and to reduce the risk of health issues like respiratory and cardiovascular problems. Individuals often stay indoors, reduce physical exertion, and use air purifiers to mitigate the effects of poor air quality. These behaviors reflect increased awareness of air pollution risks, but they also highlight the need for continued public education and efforts to improve air quality on a larger scale, reducing the reliance on personal adaptations.

Implementing stricter environmental regulations and running awareness campaigns are crucial steps in enhancing air quality and protecting public health. Stronger regulations on industries, transportation, and other major pollution sources can help reduce harmful emissions and improve air quality. Public awareness campaigns are equally important, as they educate people about the risks of air pollution, the importance of minimizing exposure, and the actions they can take to protect themselves. By informing the public and encouraging pro-environmental behaviors, these campaigns can foster a shift toward more sustainable practices, such as using public transport and conserving energy. A combination of stringent regulations and robust public education is essential to creating cleaner, healthier environments and ensuring lasting improvements in air quality.

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